

CLARA DIMARCO

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RESEARCH INTERESTS

I believe education is the most reliable way to change a child's trajectory, and my work is an effort to make that path more accessible. My research builds and evaluates AI-driven learning systems, including intelligent tutors for propositional logic, formal verification methods for LLM-generated pedagogical content, and adaptive interfaces for learners with disabilities, because the systems placed in front of students determine who gets to benefit from them. This is the same conviction behind my K–12 AI-literacy and accessible-computing outreach, and it is why I evaluate every system I build, in a classroom, a research lab, or a state infrastructure project, by whether it expands who can participate. My training in mathematics and logic gave me the formal tools to pursue this; AI-education research gave it a research question.

PUBLICATIONS

Journal Articles

- 1 Tithi, S.D., Ramesh, A.K., **DiMarco, C.**, Tian, X., Alam, N., Fazeli, K., & Barnes, T. (2025). The Promise and Limits of LLMs in Constructing Proofs and Hints for Logic Problems in Intelligent Tutoring Systems. *Computers and Education: Artificial Intelligence*.

Conference Proceedings

- 2 Tian, X., Rajapaksha, Y., Limke, A., **DiMarco, C.**, Dobar, E.B., Hill, M., Payton, J., & Barnes, T. (2026). AI Scholars Program: Scaling AI Literacy Through K–12 Outreach. *Proceedings of the AAAI Conference on Artificial Intelligence (EAAI Track)*, 40.

EDUCATION

B.S. Computer Science, Minor in Business Administration

Expected Dec. 2026

North Carolina State University | GPA: 3.84

Honors: Grand Challenges Scholars Program Scholar, Advance Personalized Learning concentration | STARS AI Scholars

Relevant Coursework: Software Engineering, Software Project Management, Data Structures, Ethics in Computing, C

Study Abroad | Computer Science Coursework | *The Hong Kong Polytechnic University*

Spring 2026

B.S. Mathematics, Magna Cum Laude | *University at Albany, SUNY*

May 2022

HCI & LEARNING SCIENCES RESEARCH

Mobile App Developer & Researcher

Jan. – Apr. 2025

Game2Learn Lab GCSP REU (PI: Dr. Tiffany Barnes), NC State University

- Developed and prototyped Proof-42, a cross-platform mobile tutoring app (React Native, Tailwind CSS, Supabase) translating AIED research on logic scaffolding into an interactive, adaptive tool.
- Conducted iterative peer usability testing to refine navigation and interactive logic elements.
- Co-authored the resulting manuscript (Tithi et al., 2025), credited with conceptualization, methodology, validation, and writing (original draft and review & editing).

AI & Logic Systems Researcher

May – Aug. 2024

Game2Learn Lab GCSP REU (PI: Dr. Tiffany Barnes), NC State University

- Developed a Disjunctive Normal Form (DNF)-based verification algorithm and a custom GPT-4 prompt-engineering technique ("bidirectional prompting") to formally verify the mathematical correctness of LLM-generated pedagogical output.

HUMAN CENTERED SYSTEMS ENGINEERING

Software Engineer, Senior Design Capstone

Aug. 2026 – Present

Rapid Post-Earthquake Bridge Assessment Tool | NC State CCEE Earthquake Research Team & AKDOT

- Engineered a full stack earthquake response tool (Python, FastAPI, SQLAlchemy, MySQL, Docker, Nginx, React, TypeScript, Tailwind CSS) used by AKDOT engineers to interpret and act on bridge integrity data after seismic events.
- Translated complex structural-assessment algorithms from legacy MATLAB models into Python, integrating real-time USGS earthquake data and ground-motion models to automate damage prediction across Alaska's bridge inventory.
- Contributed to the project's ethical design review, evaluating how non-expert field engineers would interpret and trust an automated inspection-priority ranking.

Software Engineer

Jun. – Aug. 2025

NASA Jet Propulsion Laboratory, Pasadena, CA

- Engineered a JSON-driven configuration web interface that unified fragmented multi-system workflows into a scalable environment for Deep Space Network subsystem engineering teams.
- Automated backend file workflows using Python and Perl CGI scripts, removing manual terminal generation tasks for cross-functional operators.

Systems Engineer
NASA L'SPACE Mission Concept Academy | Interdisciplinary Lunar Mission Design

May – Aug. 2024

- Co-designed a 199-page lunar water-ice detection mission design with a cross-disciplinary, nationwide student team, defining system requirements, a Science Traceability Matrix, and a Failure Modes and Effects Analysis (FMEA).

TEACHING, OUTREACH & INCLUSIVE DESIGN

Teaching Assistant, Introduction to Computing (OOP & Java)
NC State University

Aug. – Dec. 2024

- Mentored 100+ undergraduates in object-oriented programming, algorithm analysis, and debugging; held weekly office hours and graded assignments with structured, individualized feedback.

STARS AI Scholar: K–12 AI-Literacy Outreach
STARS Computing Corps, NC State University

Dec. 2023 – Sep. 2025

- Co-designed and field-tested an equitable AI/robotics curriculum for 200+ K–12 students with disabilities to evaluate scalable methods for broadening participation in computing.
- Developed and deployed accessible AI-robotics-instructional strategies across a 5th-grade pilot cohort, co-authored a peer-reviewed publication on scalable framework deployment (*Tian et al., AAAI/EAAI 2026*).

VENTURE DEVELOPMENT & LEADERSHIP

Venture Development & Research Lead
Associated Credit Management, Hong Kong SAR

May – Jul. 2026

- Executed instructional-design market analyses and user-retention profiling of regional edtech environments to provide strategic direction for new consumer ventures.
- Synced structural platform features (AI valuation metrics, escrow protocols) with regional regulatory reporting criteria.

Treasurer & Fundraising Chair
Grand Challenges Student Organization, NC State University

Aug. 2024 – Dec. 2025

- Restructured financial infrastructure to transition the cross-disciplinary engineering organization into an official, self-sustaining nonprofit entity.

HONORS & AWARDS

- Robert E. Gross / Lockheed Aircraft Corporation Scholarship, NC State College of Engineering (2025)
- STARS Award, STARS Computing Corps (2025)
- Connections Award, ASA DataFest (2025)
- Dean’s List, NC State University (2023)
- Grand Challenges Scholars Program Scholar, Advance Personalized Learning concentration (2023–2026)
- Magna Cum Laude, University at Albany, SUNY (2022)

RESEARCH ETHICS & TRAINING

- CITI Program certifications in Human Subjects Research and the Responsible Conduct of Research (2024–2025)
- Developing Cultural Competence certification (2024)

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, SQL, HTML, CSS, C, Perl

AI/ML: LLMs, GPT-4, Prompt Engineering

Tools & Databases: Figma, Git, REST APIs, MySQL, Supabase, Docker, Nginx, USGS API, Postman, Apache, Expo Go

Frameworks: React, React Native, Next.js, Tailwind CSS, FastAPI, SQLAlchemy, Pandas, Spring, Hibernate, jQuery, Bootstrap